

- Q.26 Explain the factors affecting sag in an over head line.  
 Q.27 Explain any two methods of improving string efficiency.

Define the following terms:

- a) Earth wire                      b) Shack insulator  
 c) Guard wire                      d) Lighting arrestor
- Q.28 Explain briefly the skin effect in a transmission line. On what factors does it depend?
- Q.29 Write down the advantages and disadvantages of outdoor substation over indoor substation.
- Q.30 Write a short note on pole mounted substations.
- Q.31 What are the desirable characteristics of Tariff?
- Q.32 What are the advantages of good power factor?
- Q.33 Define the following:  
 a) Power factor                      b) Voltage regulation
- Q.34 What are the cause of low power factor?
- Q.35 What do you mean by Ferranti effect? How can you reduce it?

### SECTION-D

**Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Explain the working of a Nuclear power plant with its schematic diagram.
- Q.37 a) Explain the types of faults in underground cables.  
 b) Explain Murray Loop Test and varley loop test for finding location of faults in under ground cables
- Q.38 Explain corona, factors influencing it and methods of reducing corona.

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### 5th Sem / Electrical Engineering/Elect & Eltx Engg. Sub : Electrical Power-I/Power-I (G.T. & DEP)

Time : 3Hrs.

M.M. : 100

### SECTION-A

**Note:** Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Which of the following plants will have the highest capital cost  
 a) Nuclear power plant    b) Diesel power plant  
 c) Thermal power plant    d) None of these
- Q.2 Load factor is the ratio of  
 a) Avg. Demand/Maxim Demand  
 b) Maxim. Demand/Avg. Demand  
 c) Avg. Demand/Plant Capacity  
 d) None of these
- Q.3 The effect of corona is  
 a) Increased energy loss  
 b) Increased reactance  
 c) Increased inductance  
 d) All of these
- Q.4 The main criterion for selection of the size of a distributor for radial distribution system is  
 a) Voltage drop                      b) Corona Loss  
 c) Temperature rise                      d) Capital cost
- Q.5 The power factor of leading power factor is  
 a) Leading power factor  
 b) Leading power factor

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- c) Unity power factor  
d) None of these
- Q.6 The type of insulator employed in distributions lines are  
a) Pin type                      b) Shack type  
c) Strain type                  d) Suspension type
- Q.7 It is desirable that the voltage regulation of the transmission line should be  
a) High                          b) Stable  
c) Low                          d) Very high
- Q.8 The transmission lines which feed different sub-stations represent  
a) Primary transmission  
b) Secondary transmission  
c) Primary distribution  
d) Secondary distribution
- Q.9 Step down substations are installed near  
a) Generating station  
b) Distribution sub-station  
c) Transmitting station  
d) Consumer's premises
- Q.10 The volume of copper required for an A.C. transmission line is inversely proportional to  
a) Current                      b) Voltage  
c) Power factor              d) Both B & C

### SECTION-B

**Note:** Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Define Primary Transmission.
- Q.12 The variable load which occurs over and above the base load is called \_\_\_\_\_.

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- Q.13 With the increase in voltage level of transmission line, Loss of the transmission lines gets decreased. (True / False)
- Q.14 Corona loss is minimized, if the size of the conductor is \_\_\_\_\_.
- Q.15 Demand factor in a power system is always \_\_\_\_\_ than unity.
- Q.16 Overhead system is more safer than underground system. (T/F)
- Q.17 The service mains connects the feeder and distributor. T/F
- Q.18 Murray Loop test is used to find grand fault in multi core cable )T/F)
- Q.19 Define secondary Transmission.
- Q.20 Electrical Power system with low power factor tends to draw more current from the supply (T/F).

### SECTION-C

**Note:** Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Define the terms  
a) Demand factor              b) Average load  
c) Load factor                d) Diversity factor
- Q.22 Write down the difference between underground system and over head system.
- Q.23 How does the proximity effect affects the resistance of the conductor?
- Q.24 Describe the various methods for reducing corona effect in an overhead line.
- Q.25 Differentiate between Base load and peak load power plants.

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